Intervals and absolute value equations

Polynomial-size vertex test for inverse M-matrix intervals

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Rating rationale: Replacing an exponential interval test by a quadratic vertex family is challenging; verified inverse-positivity has specialist importance in interval matrix analysis.

Area: interval linear algebra; structured matrices

Independently reviewed resolution - 2026-09-11

Affirmative resolution. Theorem 1 proves that every interval member is inverse-M if and only if the n^2 vertices $C - D_i R D_j$ are inverse-M. These are contained in the displayed two-sign family, so the original $2n^2$ equivalence follows. The proof covers all real endpoints, every dimension, zero widths, zero entries and reducible matrices without assuming regularity.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. See the complete manuscript, independent agent review and submission record. The source archive identifies the drafts as AI-generated; authorship is recorded at the submitter's request. This is agent verification, not external human peer review or formal proof-assistant certification.

The difficulty, importance and rating rationale below are historical assessments of the original open target. The original statement and dated audits are preserved.

Problem statement

A real square matrix M is a nonsingular M-matrix if its off-diagonal entries are nonpositive, M is invertible, and M^{-1} is entrywise nonnegative. An inverse M-matrix is the inverse of such a matrix.

For every $n \geq 1$, let $L, U \in \mathbb{R}^{n\times n}$ satisfy $L \leq U$ entrywise, and set

$$\mathbb{A} = \{A : L \leq A \leq U\}, \qquad C = (L+U)/2, \qquad R = (U-L)/2.$$

For $i = 1, \ldots, n$, let $z^{(i)}$ have entry -1 in position i and entry 1 elsewhere, and write $D_i = \operatorname{diag}(z^{(i)})$.

Is the following equivalence true?

$$[\text{every } A \in \mathbb{A} \text{ is an inverse M-matrix}]$$
$$\Longleftrightarrow$$
$$\left[\begin{array}{c} C + sD_iRD_j \text{ is an inverse M-matrix} \\ \text{for every } i,j \in \{1,\ldots,n\} \text{ and } s \in \{-1,1\} \end{array}\right].$$

All entries in the interval vary independently, and degenerate intervals are allowed. The proposed test uses at most $2n^2$ matrices; repeated test matrices are harmless. This would give a small exact certificate for a structured property needed in interval inversion and verified linear-system computation.

References

Milan Hladík, *An overview of polynomially computable characteristics of special interval matrices*, in *Beyond Traditional Probabilistic Data Processing Techniques*, Springer (2020), pp. 295–310; author preprint, arXiv:1711.08732v1, §9, Conjecture 1, p. 11, with the vectors defined in Theorem 24.

Jürgen Garloff, Doaa Al-Saafin, and Mohammad Adm, *Further Matrix Classes Possessing the Interval Property*, Reliable Computing **28** (2021), 56–70, p. 64, paragraph immediately before IP 4.4.

Status check

The 2021 paper explicitly retains this conjecture while proving a different sufficient vertex family of size 2^{n-1}. On 2026-09-10, searches combining the original title, inverse M-matrices, interval property, conjecture, and 2025/2026 found no resolution. Hladík's current publication list and the later interval-property paper were checked. This is a selected-source status check, not an exhaustive citation audit.

Independent audit — 2026-09-10

The full Hladík preprint, §9, Conjecture 1, agrees with the displayed two-sign vertex family, including the definition of each $z^{(i)}$. Garloff–Al-Saafin–Adm, p. 64, expressly leaves that conjecture unresolved; its exponential family is a different test, not a resolution for a subclass here. Independent later searches for the exact conjecture and matrix class found no resolution.